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PHYLOGENY AND EVOLUTIONARY HISTORY OF DIPLOBATHRID CRINOIDS (ECHINODERMATA)

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Abstract: Order Diplobathrida is a major clade of camerate crinoids spanning the Ordovician–Mississippian, yet phylogenetic relationships have only been inferred for Ordovician taxa. This has hampered efforts to construct a comprehensive tree of life for crinoids and develop a classification scheme that adequately reflects diplobathrid evolutionary history. Here, I apply maximum parsimony and Bayesian phylogenetic approaches to the fossil record of diplobathrids to infer the largest tree of fossil crinoids to date, with over 100 genera included. Recovered trees provide a framework for evaluating the current classification of diplobathrids. Notably, previous suborder divisions are not supported, and superfamily divisions will require significant modification. Although numerous revisions are required for families, most can be retained through reassignment of genera. In addition, recovered trees were used to produce phylogeny-based estimates of diplobathrid lineage diversity. By accounting for ghost lineages, phylogeny-based richness estimates offer greater insight into diversification and extinction dynamics than traditional taxonomy-based approaches alone and provide a detailed summary of the ~150 million-year evolutionary history of Diplobathrida. This study constitutes a major step toward producing a phylogeny of the Crinoidea and documenting crinoid diversity dynamics. In addition, it will serve as a framework for subsequent phylogeny-based investigations of macroevolutionary questions.

Key words: phylogeny, Diplobathrida, Crinoidea, systematics, diversification, macroevolution.

In recent years, concerted efforts have been made to improve, or in some cases provide for the first time, inference of evolutionary relationships among fossil invertebrates. The results of these efforts serve both to inform systematic revision of fossil taxa, thereby creating classifications that reflect evolutionary history, and to provide frameworks for addressing macroevolutionary questions within a phylogenetic context. Although the phylogenetic revolution has long been underway in the field of palaeobiology, echinoderms have been at the forefront of this movement in recent years, in part because of their taxonomic diversity, abundance in the fossil record, and construction from many discrete skeletal elements, which makes them a practical group for collecting morphological data for phylogenetic analysis. As a result, major advances have been made in understanding the fossil echinoderm tree of life (e.g. Smith & Zamora 2009; Kroh & Smith 2010; Ausich *et al.* 2015; Thompson *et al.* 2015; Thuy & Stöhr 2016, 2018; Bauer *et al.* 2017; Sheffield & Sumrall 2018) that have been complemented by analyses of extant echinoderms (e.g. Rouse *et al.* 2013; Miller *et al.* 2017).

In particular, much recent research has focused on resolving the phylogeny of Palaeozoic crinoids. Crinoids have a rich, well-sampled fossil record (Foote 1995; Cole 2017) that spans over 480 million years (Ordovician–Recent) and includes over 8000 named fossil species from the Palaeozoic (Lane 1990). As a result, crinoids are useful for investigating evolutionary patterns and processes in the fossil record and have been successfully used as a model group for addressing a variety of macroevolutionary questions. Building on earlier analyses (Ausich 1998*a*, *b*), Ausich *et al.* (2015) established the evolutionary relationships among major clades of Ordovician crinoids. Subsequent investigations focused on inferring relationships of Ordovician camerates (Cole 2017) and Ordovician–Devonian pentacrinoids (Wright 2017*a*), and results from these analyses were used to inform a revised, phylogeny-based classification of Class Crinoidea (Wright *et al.* 2017). More recently, additional phylogenetic investigations have been conducted for Palaeozoic eucladids (Wright 2017*b*), Ordovician–Silurian flexibles (Wright & Toom 2017), Ordovician monobathrids (Cole & Toom 2018), Ordovician diplobathrids (Cole *et al.* 2018), disparids (Ausich 2018) and iocrinids (Lin *et al.* 2018). As a result of these combined efforts, the phylogeny of fossil crinoids is better understood than that of most Palaeozoic marine invertebrates.

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The inferred crinoid phylogeny provides a valuable framework for addressing many new questions within a phylogenetic context, including systematic revision, macroevolutionary studies using phylogenetic comparative methods, and integration with molecular phylogenies. Despite significant advances in resolving the crinoid tree of life, however, much work remains before a comprehensive phylogenetic framework is developed. Notable crinoid clades for which evolutionary relationships still remain to be addressed include: (1) diplobathrid camerates; (2) monobathrid camerates; and (3) eucladids (*sensu* Wright 2017*a*), particularly the post-Palaeozoic Articuliformes that gave rise to the crown-group articulate crinoids.

Here, I use parsimony and Bayesian approaches to infer phylogenetic relationships within the Diplobathrida, one of two orders in subclass Camerata. Previous phylogenetic investigation confirmed that both Diplobathrida and its sister clade, Monobathrida, are monophyletic groups (Cole 2017), but evolutionary relationships among post-Ordovician diplobathrids have not yet been assessed. The phylogenetic analyses presented here include all well-preserved diplobathrids (101 genera), making it the largest phylogeny inferred for fossil crinoids to date. In addition, I present diversity plots for diplobathrid crinoids using both taxonomic diversity data and phylogeny-based estimates and discuss the evolutionary trajectory of the clade. Results of this study constitute a major step toward producing a phylogeny of all crinoids and understanding crinoid diversity dynamics across different phylogenetic scales, with particular focus on evaluating suborder- and family-level classification of diplobathrids. In addition, the recovered trees will serve as a framework for subsequent high-resolution investigations of macroevolutionary questions.

DIPLOBATHRIDS: AN OVERVIEW

Evolutionary palaeoecology

Diplobathrids are a moderately diverse order of camerate crinoids with over 100 genera and 350 named species. The fossil record of diplobathrid crinoids spans about 154 million years from the Early Ordovician (Floian) to late Mississippian (Serpukhovian). Because the Tremadocian-age dicyclic camerate *Cnemecrinus fillmorensis* Guensberg & Sprinkle, 2003 has been recognized as a stem eucamerate rather than a true diplobathrid (Cole 2017), the earliest known diplobathrid is *Proexenocrinus inyoensis* Strimple & McGinnis, 1972 from the Floian of California. Like most major groups of crinoids, however, diplobathrids did not become abundant or taxonomically diverse until the Middle to Late Ordovician when they diversified as part of the Great Ordovician Biodiversification Event (GOBE; Webby *et al.* 2004).

Diplobathrids were significant constituents of the early Palaeozoic Crinoid Evolutionary Fauna (CEF; Ausich *et al.* 1994; Baumiller 1994) during the Ordovician, particularly the Sandbian–Katian (Ausich & Deline 2012; Cole 2017; Cole *et al.* 2017). Following the end-Ordovician mass extinction, diplobathrids underwent a taxonomic radiation, but the clade as a whole never again resumed its role as a major component of a CEF. Instead, monobathrids replaced diplobathrids as the dominant camerate constituents of the Middle Palaeozoic CEF (Eckert 1988; Ausich *et al.* 1994). Despite the minor contribution of diplobathrids to post-Ordovician faunas overall, certain taxa were significant constituents of faunas locally, occurring in high relative abundance. Examples include *Dimerocrinites* in the middle Silurian Much Wenlock Limestone (UK) and Rochester Shale (New York), *Thylacocrinus* in the Lower Devonian Saint-Cenere Formation (Germany, France), *Rhodocrinites* and *Cribanocrinus* in the lower Mississippian Burlington and LeGrand faunas (Iowa, Illinois, Missouri), and *Gilbertsocrinus* in the lower Mississippian Crawfordsville fauna (Indiana).

With few exceptions, diplobathrids had pinnulate arms and very dense filtration fans. Within Mississippian communities, diplobathrids had the highest density filtration fans (Ausich 1980) and similar results from Ordovician faunas suggest this may have been true of diplobathrids throughout their history (Brower 2007). Although monobathrid camerates are also pinnulate, the density of their filtration fan is typically less than that of diplobathrids and more comparable to that of 'poteriocrines' (Ausich 1980; note that poteriocrines are no longer considered to be a valid group but are broadly correlative to the articuliform cladids of Wright (2017*b*)). It has been hypothesized that the dense filtration fan of diplobathrids would have led to a narrow tolerance for environmental conditions like water turbulence and sedimentary environments, making them specialist taxa (Kammer 1985; Kammer & Ausich 1987, 2006; Holterhoff 1997), which possibly contributed to their diminished role in post-Ordovician CEFs.

Morphology and traditional classification

Like all camerates, diplobathrids possess rigidly ankylosed thecal and tegminal plates, fixed brachials and interbrachials incorporated into the calyx, a subtegminal mouth, and typically additional plates in the posterior interray (Ubaghs 1978). Diplobathrida is characterized by two circlets of plates (basals and infrabasals) below the radial circlet (Moore & Laudon 1943). The sister group to diplobathrids, order Monobathrida, is characterized by having only one circlet of plates (basals) below the radial circlet.

Building on the classifications outlined by Moore & Laudon (1943) and Ubaghs (1953), the *Treatise on Invertebrate Paleontology* divided diplobathrid genera among two suborders, three superfamilies, and 13 families (Ubaghs 1978), although some of these higher taxa have since been formally or informally dissolved (e.g. suborder Zygodiplobathrina (Brower 1975; Kelly *et al.* 1978; Kolata 1982), family Archaeocrinidae (Brower & Veinus 1974; Kolata 1982)) and others have been erected (e.g. family Callistocrinidae (Eckert & Brett 2001), family Aishacrinidae (Prokop 2010), subfamily Gilberstocrininae (Pidal 1984)). Suborders were traditionally divided based on the condition of the basal and radial circlets. The Zygodiplobathrida, which traditionally included only *Cleiocrinus* and *Spyridiocrinus*, possess basals and radials in a single circlet of 10 plates, whereas the Eudiplobathrida, which includes all other diplobathrids, have the basals and radials in separate circlets. The use of these subordinal divisions has long been questioned, however, and has largely been abandoned (Brower 1975; Kelly *et al.* 1978; Haugh 1979; Kolata 1982; Ausich 1986). The primary morphological feature traditionally used to separate diplobathrid superfamilies is the interruption of the radial circlet by the basal and interray plates, i.e. all radials separated (Rhodocrinitoidea), radials separated in the posterior interray only (Dimerocrinitoidea), or radials not separated (Nyctocrinoidea) (Fig. 1). Differentiation among families is based on more varied characters that include the number of bifurcations within the fixed brachials, position of the infrabasals, condition of the posterior interray, depression and/or plating of the regular interrays, presence/absence of median ray ridges, and calyx shape (Ubaghs 1978).

At the superfamily level, about 72% of taxa have been classified as belonging to Rhodocrinitoidea and 27% to Dimerocrinitoidea. Nyctocrinoidea is monospecific. At the family level, the majority of genera have been assigned to one of two groups: Rhodocrinitidae, which includes approximately 47% of all diplobathrid genera, or Dimerocrinitidae, which includes approximately 21% of genera. However, there has been much conjecture that the major superfamily and family groupings are polyphyletic. For example, Frest & Strimple (1981) noted that the dimerocrinitoids were often convergent on rhodocrinitoids and speculated that the two superfamilies would eventually need to be synonomized. Because both groups have been very broadly defined with few characters for diagnosis, their monophyly has long been considered dubious. Although the condition of the radial circlet is typically treated as diagnostic of higher taxa within diplobathrids, many genera within the traditional superfamilies and families have variable radial plate arrangement (Frest & Strimple 1981). As a result, it has long been suspected that different configurations in radial separation represent evolutionary grades within Diplobathrida rather than true clades, although this has not been adequately addressed in a phylogenetic context.

Previous phylogenetic work

A summary of previous hypotheses of camerate phylogeny and evolution, including both qualitative discussions (e.g. Moore 1952; Ubaghs 1978) and analytical assessments (Simms 1993; Ausich 1998*a*, *b*; Guensburg 2012), was presented by Cole and Ausich (2015). Since this summary, the phylogenetic analysis of Ordovician crinoids by Ausich *et al.* (2015) established the monophyly of subclass Camerata, formed the basis for erecting subclass

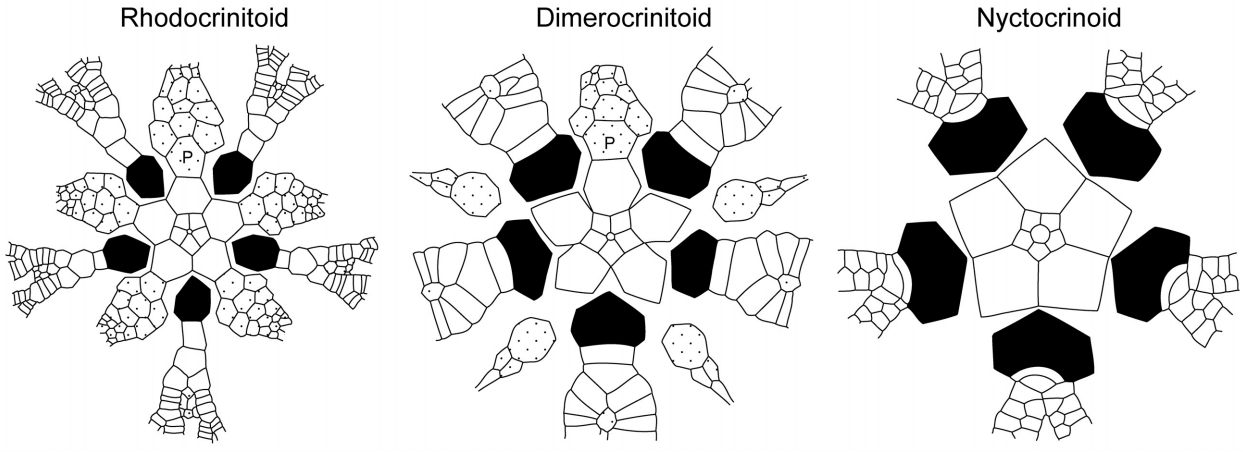


FIG. 1. Exploded plate diagrams for major calyx design patterns of diplobathrid crinoids. Separation of radial plates corresponds to the traditional delineation of superfamilies: all radials separated = rhodocrinitoids; radials separated in posterior only = dimerocrinitoids; radials not separated = nyctocrinoids. Black fill = radial plates, stipple = interray and intrabrachial plates, P = primanal. Redrawn from Ubaghs (1978).

Pentacrinoidea (Wright *et al.* 2017), and recognized subclasses Camerata and Pentacrinoidea diverged early in the evolutionary history of crinoids. Subsequently, Cole (2017) presented a phylogeny of all well-preserved Ordovician camerate genera that established the monophyly of Monobathrida and Diplobathrida. As a result of this study, superorder Eucamerata was established (Cole 2017; Wright *et al.* 2017) to accept Monobathrida and Diplobathrida, and other camerate taxa not belonging to Eucamerata were informally termed 'stem eucamerates'. Dicyclic taxa traditionally assigned to Diplobathrida that were recovered as stem eucamerates rather than true diplobathrids include *Cnemecrinus*, *Rosfacrinus* and reteocrinids *Reteocrinus* and *Quechuacrinus*. Although *Gaurocrinus* was recovered as part of the eucamerate clade, it plotted as sister to Monobathrida rather than within the diplobathrid clade (Cole 2017).

Despite work on the phylogeny of Ordovician camerates, the evolutionary reality of diplobathrid taxa and their phylogenetic relationships remain poorly understood. For example, there are only three Ordovician dimerocrinitid genera, so incorporation of post-Ordovician dimerocrinitids is necessary for evaluating the group as a whole. Likewise, although the analysis of Cole (2017) produced a well-resolved topology, support metrics were low for many deep nodes. As a result, the monophyly of many diplobathrid higher taxa remains to be tested. Accordingly, assessments of the superorder and family level divisions within diplobathrid crinoids are of particular interest in this study.

METHOD

Phylogenetic analyses

Taxon selection. To investigate relationships among diplobathrid crinoids, phylogenetic inferences were conducted at the genus level. I initially considered all 104 currently valid genera assigned to Diplobathrida (*sensu* Cole 2017) but four genera (*Xysmacrinus*, *Siderocrinus*, *Chaosocrinus* and *Monstrocrinus*) were excluded because of the lack of adequately preserved specimens. In a previous phylogenetic analysis (Cole 2017), it was unclear whether two dicyclic camerates (*Rosfacrinus* and *Gaurocrinus*) were true monobathrids, true diplobathrids or stem ecuamerates. For this reason, *Rosfacrinus* and *Gaurocrinus* were included along with four monobathrids (*Celtocrinus*, *Glyptocrinus*, *Pycnocrinus* and *Abludoglyptocrinus*) and two stem eucamerates (*Eknomocrinus* and *Reteocrinus*) so that the uncertain relationships of *Rosfacrinus* and *Gaurocrinus* to other camerate groups could be reassessed. The monobathrids and stem eucamerate genera were selected based on their stratigraphic position (*Celtocrinus* is the oldest known monobathrid and *Eknomocrinus* is the oldest stem eucamerate) and broad representation of early monobathrid and stem eucamerate morphology. Thus, a total of 108 genera were coded for phylogenetic analysis.

For most genera, a single representative species was coded for inclusion in the phylogenetic analysis. The species selected were typically the type species except in cases where a non-type species preserved more characters than the type. Three diplobathrid genera have notably high species richness and morphological disparity: *Rhodocrinites* (40 species), *Dimerocrinites* (36 species), and *Gilbertsocrinus* (30 species). For this reason, two species were coded for each of these three genera to capture better their overall morphological variability.

Analyses were conducted both with and without inclusion of the unusual Middle Ordovician (Katian) genus, *Cleiocrinus*, because of its questionable affinity with Diplobathrida. *Cleiocrinus* was assigned first to the Crotalicrinidae (Zittel 1879) and then to the flexible family Icthyocrinidae (Wachsmuth & Springer 1886), but was later transferred to Camerata on the basis of its fixed brachials incorporated into the calyx and to Diplobathrida on the basis of its two circlets of plates in addition to the radial circlet (Springer 1905). However, *Cleiocrinus* has many features that are not typical of diplobathrids, including radials and basal plates in a single circlet of 10 plates, a very high calyx (height more than twice its width), pores along calyx plate sutures, and no fixed interray plates (Springer 1905). Notably, the presence of interray plates is considered a synapomorphy of Camerata (Ubaghs 1978; Cole 2017), which calls into question the classification of *Cleiocrinus* as a camerate. This presents several complications for phylogenetic analysis. First, the distinct possibility that *Cleiocrinus* is neither a diplobathrid nor a camerate could result in a spurious tree topology if included in a phylogenetic analysis of diplobathrids. Second, even if *Cleiocrinus* is a camerate, it is very likely that its abundance of autapomorphies and paucity of characters shared with other diplobathrids would lead to erroneous relationships through long branch attraction. Despite these complications, however, I deemed it necessary to run an analysis with *Cleiocrinus* included in order to assess the legitimacy of the diplobathrid suborder Zygodiplobathrina, to which *Cleiocrinus* was formerly assigned. Zygodiplobathrina was erected by Ubaghs (1953) to accept *Cleiocrinus* and *Spyridiocrinus*, which share the arrangement of basal and radial plates in a single circlet of 10 plates; all other diplobathrids were assigned to suborder Eudiplobathrina.

Character coding. Representative species were coded using museum specimens available in the collections of the

Smithsonian Institution National Museum of Natural History (Washington, DC), the Field Museum (Chicago, IL), the Natural History Museum (London, UK), the Lapworth Museum (Birmingham, UK), and Trinity College (Dublin, Ireland). Primary taxonomic literature was used to supplement coding from specimens and in instances where museum specimens were inaccessible (Cole 2018, suppl. table 1). Species were coded as a proxy for each genus using 135 discrete, morphological characters, 123 of which proved to be parsimony informative (Cole 2018, suppl. table 2). Characters were selected to capture morphological variation among diplobathrid crinoids and are inferred to represent homologous structures among sampled taxa (Ausich 1996; Kammer *et al.* 2013; Wright 2015).

Outgroup selection. Building upon well-supported clades recovered from the analysis of Cole (2017), *Eknomocrinus* was designated the outgroup for all phylogenetic analyses with a simple additional constraint imposed to make *Reteocrinus* sister to Eucamerata and constrain established members of Monobathrida and Diplobathrida to their respective clades (Cole 2018, suppl. fig. 1). Within Diplobathrida, the early Ordovician (Floian) genus *Proexenocrinus* was constrained as sister to all other diplobathrids, effectively designating *Proexenocrinus* as the immediate outgroup to Diplobathrida. Proexenocrinus was used as the outgroup because it is the oldest known diplobathrid by approximately 10 million years. *Rosfacrinus* and *Gaurocrinus* were not included in the constraint tree so their positions would be free to vary in the phylogenetic analyses.

Phylogenetic inference. Parsimony analyses were first conducted in PAUP* v. 4.0a147 (Swofford 2003) using heuristic searches with 10 000 random addition sequences. Tree bisection reconnection was used for the branch-swapping algorithm with no reconnection limit, and branches with a maximum length of zero were collapsed. All characters were left unordered and equally weighted. This analysis was run twice, once with and once without *Cleiocrinus* included.

To assess robustness of the recovered trees, jackknife, bootstrap, and Bremer support values were calculated in PAUP*. One hundred bootstrap replicates were run with a full heuristic search of 10 replicates and all parsimony-informative characters resampled. One hundred jackknife replicates were run with a full heuristic search of 10 replicates and 20% of characters resampled. Average consistency index (CI) and retention index (RI) values were recorded for the characters in the recovered trees.

Finally, a Bayesian phylogenetic analysis was conducted to compare the results of different phylogenetic approaches in diplobathrid phylogeny reconstruction. Taxon sampling and outgroup constraints imposed were the same as for the parsimony analyses. The analysis was run in MrBayes 3.2.6 (Ronquist *et al.* 2012) using Markov chain Monte Carlo over 5 million generations with a sampling frequency of 5000. Where recovered clades were consistent between the Bayesian majority rule consensus tree and the parsimony strict consensus tree, posterior probabilities were recorded as an additional support metric.

Additional analyses

To assess character support for recovered clades and identify patterns of morphological evolution, character analyses were conducted in Mesquite v. 3.5 (Maddison & Maddison 2018). Parsimony reconstruction of ancestral states was used to trace character state transitions across the tree and to identify characters shared among recovered groups that might be useful for diagnoses of higher taxa.

Using the R package `strap` (Bell & Lloyd 2015), the parsimony strict consensus tree was plotted against the observed stratigraphic ranges of sampled diplobathrid genera to produce a time-scaled phylogeny. Stratigraphic ranges of genera used to time-scale the tree were tabulated from Webster (2003) and updated to include new taxa (Webster & Webster 2014) and match current divisions of the International Chronostratigraphic Chart. To assess the fit of the recovered tree to observed stratigraphic ranges of the constituent taxa, `strap` was also employed to calculate stratigraphic congruence metrics by comparing the recovered tree to a set of randomly generated tree topologies based on observed ranges of the taxa included in the tree. Stratigraphic congruence measures calculated were the Stratigraphic Consistency Index (SCI; Huelsenbeck 1994), Relative Completeness Index (RCI; Benton & Storrs 1994), the Gap Excess Ratio (GER; Wills 1999) and Manhattan Stratigraphic Measure (MSM*; Siddall 1998; Pol & Norell 2001). For each measure, the calculated value was compared to a null model of 1000 trees generated through Monte Carlo simulation, and a *p*-value was calculated for each of the observed metrics.

Finally, to better understand the evolutionary history of diplobathrid diversification, two different plots of taxonomic diversity were generated. First, genus and species-level diversity were calculated using the range-through method. Second, phylogeny-based estimates of genus diversity trees were investigated. Each of the recovered MPTs was stochastically time-scaled 10 times using the cal3 method (Bapst 2013), and the resulting sample of 150 time-scaled trees was used to plot median diversity using the `multiDiv ()` function in the R package

`paleotree` (Bapst 2012). This approach to calculating diversity includes both sampled taxa and all ghost lineages implied by the recovered phylogenetic trees. To maintain time bins of approximately similar durations for both analyses, geological stages were used as time bins for the Ordovician, Devonian and Mississippian, and epochs were used for the Silurian.

RESULTS

Phylogenetic inference

Comparison of sampling and phylogenetic approaches. The parsimony analysis excluding *Cleiocrinus* recovered 15 most-parsimonious trees of 1598 steps in length with CI and RI values of 0.217 and 0.46, respectively (Fig. 2). The parsimony analysis including *Cleiocrinus* recovered 41 most-parsimonious trees of 1621 steps in length with CI and RI values of 0.215 and 0.46, respectively (Cole 2018, suppl. fig. 2). Strict consensus trees resulting from the two analyses are largely congruent. Inclusion of *Cleiocrinus* did not recover Zygodiplobathrina as an evolutionarily meaningful group. Although both *Spyridiocrinus* and *Cleiocrinus* fall within the same major clade, *Spyridiocrinus* plots as most closely related to a subclade comprised of rhodocrinitoids (including anthracocrinids and rhodocrinitids) and one gazacrinid, whereas *Cleiocrinus* plots as sister to the rhodocrinitids *Paulocrinus* and *Rheocrinus*. Interestingly, *Rheocrinus* is one of only a few diplobathrids known to have the radials in contact with the infrabasals, resulting in a circlet of 10 plates similar to that of *Cleiocrinus*. Despite this condition, the morphology of *Rheocrinus* is like that of other eudiplobathrines in all other respects, and the genus has traditionally been assigned to family Rhodocrinitidae (Haugh 1979). Although it is possible that this recovered relationship between *Rheocrinus* and *Cleiocrinus* is real, the unusual morphology of *Cleiocrinus* still raises concerns about long branch attraction and spurious topologies, so all subsequent discussions will focus on the results of analyses that excluded *Cleiocrinus*.

The strict consensus tree resulting from parsimony analysis is well-resolved with only small, localized polytomies. The 50% majority rule consensus tree recovered from the Bayesian analysis is more poorly resolved, with structure restricted to more nested clades within the tree. The resolved portions of the Bayesian consensus tree largely correspond to portions of the parsimony strict consensus tree for which support metrics were recovered.

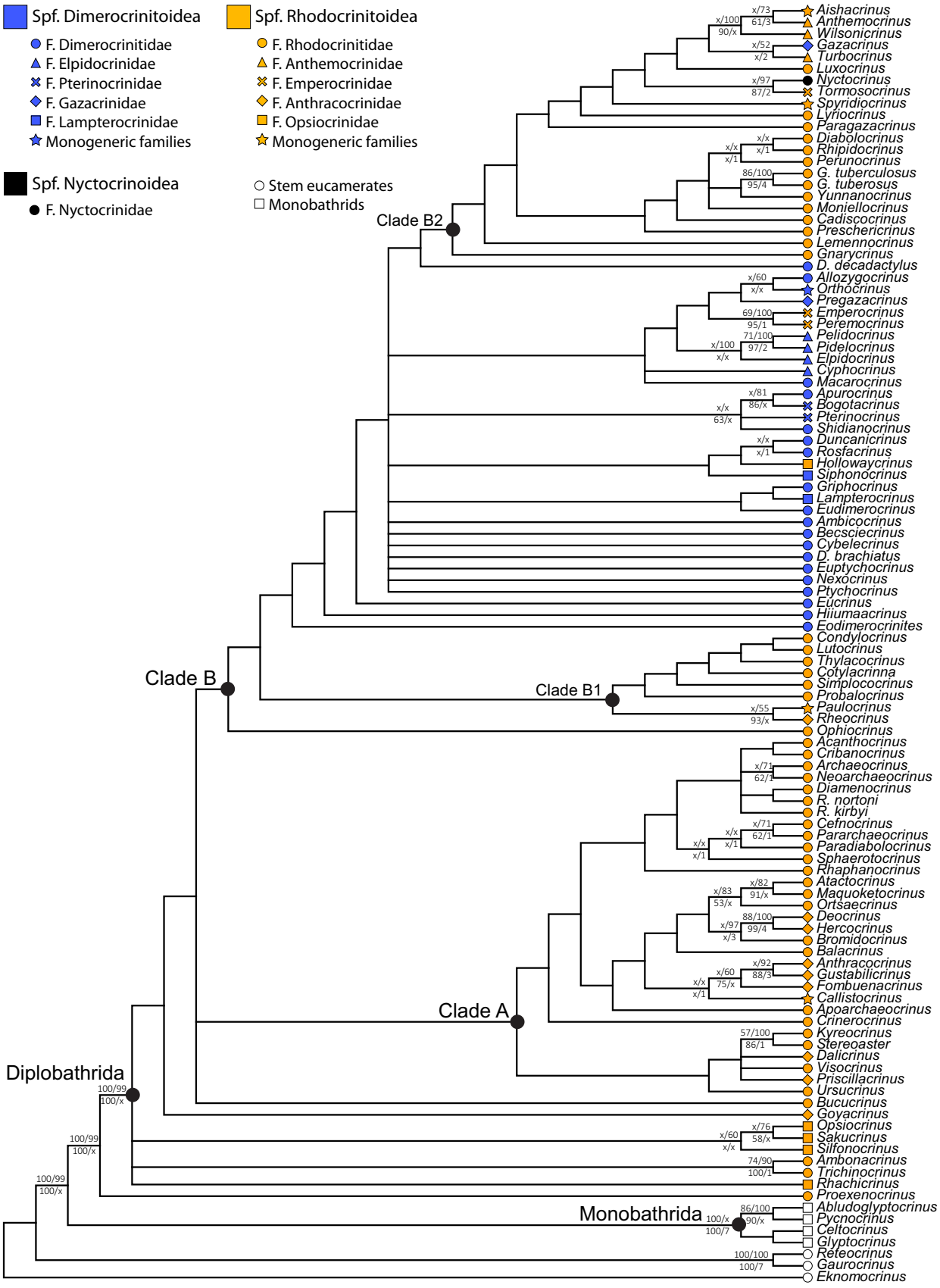
Tree topology and support. With high support in both parsimony and Bayesian analyses, *Gaurocrinus* plotted as sister to the stem eucamerate *Reteocrinus*, and *Rosfacrinus* plotted within Diplobathrida (Fig. 2). In the strict consensus tree, eight diplobathrids plot as stem lineages near the base of the tree with some forming small clades. All of these stem taxa have traditionally been assigned to superfamily Rhodocrinitoidea, and range in age from Middle Ordovician to Middle Devonian (Fig. 3). Three genera belonging to family Opsiocrinidae (*Sakucrinus*, *Silfonocrinus* and *Opsiocrinus*) form a clade. The remaining stem lineage taxa are currently assigned to family Rhodocrinitidae.

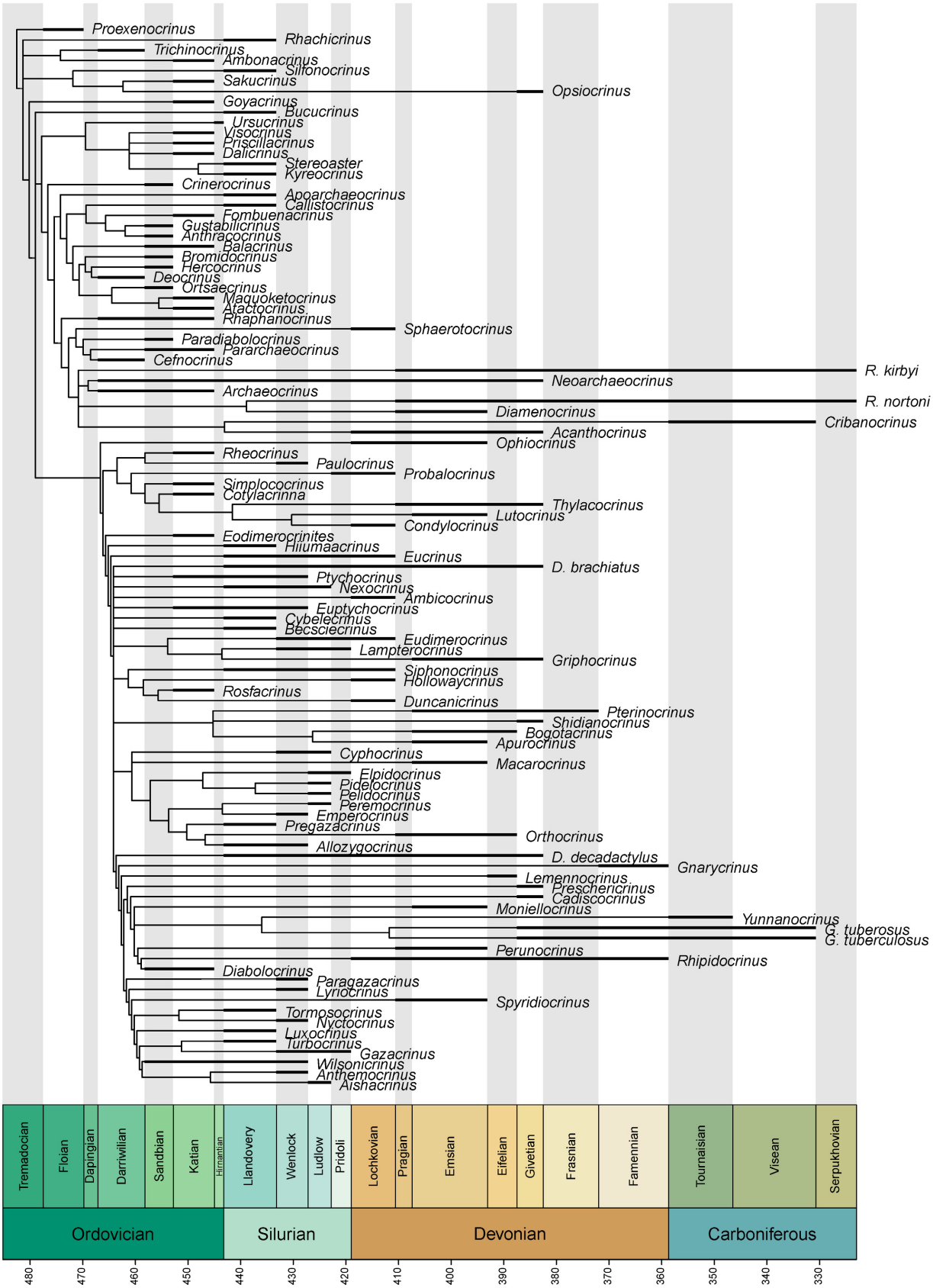
All other diplobathrids fall within two large clades ('Clade A' and 'Clade B'; Fig. 2). These two groups are broadly divided by age. The taxa comprising Clade A are predominantly Ordovician and early Silurian in age, with only six genera extending into the Devonian and/or Mississippian (Fig. 3). By contrast, Clade B is predominantly comprised of taxa that are Silurian to Mississippian in age with only nine taxa originating in the Ordovician.

Clade A is the smaller of the two groups with 30 genera recovered in the parsimony strict consensus tree and 31 genera recovered in the Bayesian 50% majority rule tree (*Bucucrinus* plots along a polytomy with Clades A and B in the strict consensus but is recovered as a member of clade A in the 50% majority rule tree). In terms of the traditional classification at the superfamily level, Clade A is comprised entirely of taxa belonging to Rhodocrinitoidea. At the family level, Clade A encompasses all genera traditionally assigned to Anthracocrinidae, approximately 62% of genera belonging to Rhodocrinitidae, and the single genus belonging to Callistocrinidae. Notably, both species of *Rhodocrinites* that were included in the analysis are recovered in Clade A as closely related species. In addition, all of the rhodocrinitid genera formerly assigned to Archaeocrinidae (*Archaeocrinus*, *Neoarchaeocrinus*, *Pararchaeocrinus*, *Rhapanocrinus* and *Balacrinus*) are recovered in Clade A.

Clade B is comprised of 61 genera, including all those traditionally assigned to superfamily Dimerocrinitoidea, the remaining members of Rhodocrinitoidea, and the single genus assigned to Nyctocrinoidea. Broadly, Clade B

FIG. 2. Strict consensus of 15 most-parsimonious trees recovered from parsimony analysis of diplobathrid crinoids. Values at nodes represent support metrics compiled from parsimony and Bayesian phylogenetic analyses: bootstrap/jackknife (above), posterior probability/Bremer support (below), 'x' given where significant support value were not recovered. Symbols at terminal tips represent the traditional classification of genera at the superfamily (Spf.) and family (F.) levels. *Genus abbreviations*: *R.*, *Rhodocrinites*; *D.*, *Dimerocrinites*; *G.*, *Gilbertsocrinus*. Colour online.


Spf. Dimerocrinitoidea
F. Dimerocrinitidae
F. Elpidocrinidae
F. Pterinocrinidae
F. Gazacrinidae
F. Lampterocrinidae
Monogeneric families
Spf. Rhodocrinitoidea
F. Rhodocrinitidae
F. Anthemocrinidae
F. Emperocrinidae
F. Anthracocrinidae
F. Opsiocrinidae
Monogeneric families
Spf. Nyctocrinoidea
F. Nyctocrinidae
Stem eucamerates
Monobathrids
Clade B2
Clade B
Clade B1
Clade A
Diplobathrida
Monobathrida
Aishacrinus
Anthemocrinus
Wilsonicrinus
Gazacrinus
Turbocrinus
Luxocrinus
Nyctocrinus
Tormosocrinus
Spyridiocrinus
Lyriocrinus
Paragazacrinus
Diabolocrinus
Rhipidocrinus
Perunocrinus
G. tuberculosus
G. tuberosus
Yunnanocrinus
Moniellocrinus
Cadiscocrinus
Preschericrinus
Lemennocrinus
Gnarycrinus
D. decadactylus
Allozygocrinus
Orthocrinus
Pregazacrinus
Emperocrinus
Peremocrinus
Pelidocrinus
Pidelocrinus
Elpidocrinus
Cyphocrinus
Macarocrinus
Apurocrinus
Bogotacrinus
Pterinocrinus
Shidianocrinus
Duncanicrinus
Rosfacrinus
Hollowaycrinus
Siphonocrinus
Griphocrinus
Lampterocrinus
Eudimerocrinus
Ambicocrinus
Becsciecrinus
Cybelecrinus
D. brachiatus
Euptychocrinus
Nexocrinus
Ptychocrinus
Eucrinus
Hiiumaacrinus
Eodimerocrinites
Condylocrinus
Lutocrinus
Thylacocrinus
Cotylacrinna
Simplococrinus
Probalocrinus
Paulocrinus
Rheocrinus
Ophiocrinus
Acanthocrinus
Cribanocrinus
Archaeocrinus
Neoarchaeocrinus
Diamenocrinus
R. nortoni
R. kirbyi
Cefnocrinus
Pararchaeocrinus
Paradiabolocrinus
Sphaerotocrinus
Rhaphanocrinus
Atactocrinus
Maquoketocrinus
Ortsaecrinus
Deocrinus
Hercocrinus
Bromidocrinus
Balacrinus
Anthracocrinus
Gustabilicrinus
Fombuenacrinus
Callistocrinus
Apoarchaeocrinus
Crinerocrinus
Kyreocrinus
Stereoaster
Dalicrinus
Visocrinus
Priscillacrinus
Ursucrinus
Bucucrinus
Goyacrinus
Opsiocrinus
Sakucrinus
Silfonocrinus
Ambonacrinus
Trichinocrinus
Rhachicrinus
Proexenocrinus
Abludoglyptocrinus
Pycnocrinus
Celtocrinus
Glyptocrinus
Reteocrinus
Gaurocrinus
Eknomocrinus


Proexenocrinus
Rhachicrinus
Trichinocrinus
Ambonacrinus
Siltonocrinus
Sakucrinus
Opsiocrinus
Goyacrinus
Bucucrinus
Ursucrinus
Visocrinus
Priscillacrinus
Dalicrinus
Stereoaster
Kyreocrinus
Crinerocrinus
Apoarchaeocrinus
Callistocrinus
Fombuenacrinus
Gustabilicrinus
Anthracocrinus
Balacrinus
Bromidocrinus
Hercocrinus
Deocrinus
Ortsaecrinus
Maquoketocrinus
Atactocrinus
Rhaphanocrinus
Sphaerotocrinus
Paradiabolocrinus
Pararchaeocrinus
Cefnocrinus
R. kirbyi
Neoarchaeocrinus
Archaeocrinus
R. nortoni
Diamenocrinus
Cribanocrinus
Acanthocrinus
Ophiocrinus
Rheocrinus
Paulocrinus
Probalocrinus
Simplococrinus
Cotylacrinna
Thylacocrinus
Lutocrinus
Condylocrinus
Eodimerocrinites
Hiiumaacrinus
Eucrinus
D. brachiatus
Ptychocrinus
Nexocrinus
Ambicocrinus
Euptychocrinus
Cybelecrinus
Becsciecrinus
Eudimerocrinus
Lampterocrinus
Griphocrinus
Siphonocrinus
Hollowaycrinus
Rosfacrinus
Duncanicrinus
Pterinocrinus
Shidianocrinus
Bogotacrinus
Apurocrinus
Cyphocrinus
Macarocrinus
Elpidocrinus
Pidelocrinus
Pelidocrinus
Peremocrinus
Emperocrinus
Pregazacrinus
Orthocrinus
Allozygocrinus
D. decadactylus
Gnarycrinus
Lemennocrinus
Preschericrinus
Cadiscocrinus
Moniellocrinus
Yunnanocrinus
G. tuberosus
G. tuberculosus
Perunocrinus
Rhipidocrinus
Diabolocrinus
Paragazacrinus
Lyriocrinus
Spyridiocrinus
Tormosocrinus
Nyctocrinus
Luxocrinus
Turbocrinus
Gazacrinus
Wilsonicrinus
Anthemocrinus
Aishacrinus
Tremadocian
Floian
Dapingian
Darriwilian
Sandbian
Katian
Hirnantian
Llandovery
Wenlock
Ludlow
Pridoli
Lochkovian
Pragian
Emsian
Eifelian
Givetian
Frasnian
Famennian
Tournaisian
Visean
Serpukhovian
Ordovician
Silurian
Devonian
Carboniferous
480
470
460
450
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can be divided into three parts. First, a subclade of eight taxa is positioned basally within Clade B that is comprised of rhodocrinitoid genera, most of which are currently assigned to Rhodocrinitidae (Clade B1; Fig. 2). Second, all but one genus traditionally assigned to superfamily Dimerocrinitoidea fall along a grade within Clade B. This dimerocrinitoid grade is only moderately resolved in the strict consensus, although it is divided into two major groups in the 50% majority rule tree. Three rhodocrinitoid taxa also fall within this grade: the opsiocrinid, *Hollowaycrinus*, and two emperocrinids, *Emperocrinus* and *Peremocrinus*. Finally, another subclade of 21 genera is recovered that is more nested within Clade B (Clade B2; Fig. 2). This group is comprised of rhodocrinitoids with the exception of one dimerocrinitoid, *Gazacrinus*, and the sole member of Nyctocrinoidea. In terms of traditional family-level classification, the rhodocrinitoids in this subclade include all members of Anthemocrinidae, a single emperocrinid, genera from two monospecific families, and 13 rhodocrinitids. All members of subfamily Gilbertsocrininae are also recovered as part of this subclade.

All stratigraphic congruence metrics calculated for the strict consensus tree were statistically significant, with the exception of the SCI, the most conservative metric, indicating the fit of the recovered tree to stratigraphy based on the observed stratigraphic ranges of taxa was generally significantly higher than for randomly generated trees (Table 1; Figs 3, 4). In addition, high support was recovered for the separation and branching order of stem eucamerates, Monobathrida and Diplobathrida. Within Diplobathrida, however, Bremer support and other support metrics greater than 50% (bootstrap, jackknife and posterior probability) were recovered only for more nested nodes within the tree. Although many of these support values are reasonably high for closely related genera, no support values $\geq 50\%$ were recovered for deeper nodes, including those representing clades A and B.

TABLE 1. Stratigraphic congruence metrics for the strict consensus tree recovered from parsimony analysis.

Stratigraphic congruence measure	Calculated value	*p*-value
Stratigraphic Consistency Index	0.472	0.13
Relative Completeness Index	−9.761	6.4×10^{-9}
Gap Excess Ratio	0.617	6.3×10^{-9}
Manhattan Stratigraphic Measure*	0.066	2.2×10^{-13}

Evolutionary diversification

The two approaches employed for investigating diversification history of diplobathrid crinoids at the genus level produce similar overall patterns, although notable differences are present (Fig. 5). Because the phylogenetic approach tabulates the number of genera that must have been present in a time bin given the recovered tree topology, it produces estimates of genus richness that are much higher than those recovered in the taxonomic approach, with maximal richness being nearly triple that of any other sampled interval. Despite uncertainties that exist for the recovered tree topologies, results from the phylogenetic approach are probably more realistic than range-through counts of sampled taxa because of incompleteness of the fossil record.

Broadly, genus-level diversity calculated from range-through data of sampled taxa records a rapid diversification that began during the Middle Ordovician and continued through the Katian when maximal Ordovician genus richness was reached. Following a major drop in diversity at the end-Ordovician, a second major radiation occurred, which led to peak diplobathrid diversity during the Llandovery. After another sharp decline during the late Silurian, genus diversity then plateaued through the Early and Middle Devonian before decreasing rapidly during the Middle–Late Devonian and then more gradually into the Mississippian. By contrast, the phylogenetic approach of estimating diplobathrid diversity records an explosive radiation that began in the Early Ordovician and led to peak diplobathrid genus richness during the Katian. Following a relatively minor drop in diversity at the end-Ordovician, diplobathrids reached a second peak during the Llandovery, declined steadily through the Silurian to Middle Devonian, and then persisted at low diversity through the Late Devonian and Carboniferous until their eventual extinction.

Major similarities in diversification patterns recovered by both approaches include explosive diversification during the Ordovician, peaks in generic richness during the Katian and Llandovery, sharp decreases in diversity during the end-Ordovician and at the Givetian–Frasnian boundary, and a gradual decrease through the Late Devonian and Carboniferous until the ultimate demise of the clade. However, in the phylogenetic approach, the Ordovician diversification begins earlier (Floian vs Dapingian) and the end-Ordovician decrease is less severe. In addition, sampled diversity is highest during the Silurian peak, whereas the phylogenetic estimate of diversity is highest during the Katian. Finally, rather than recovering

FIG. 3. Time-scaled phylogeny of diplobathrid crinoids. Tree topology is the strict consensus of 15 most-parsimonious trees. Branches are scaled to time using the equal method for visualization purposes; observed stratigraphic ranges are shown as thick branches and inferred ghost lineages are shown as thin branches. *Genus abbreviations*: *R.*, *Rhodocrinites*; *D.*, *Dimerocrinites*; *G.*, *Gilbertsocrinus*. Colour online.

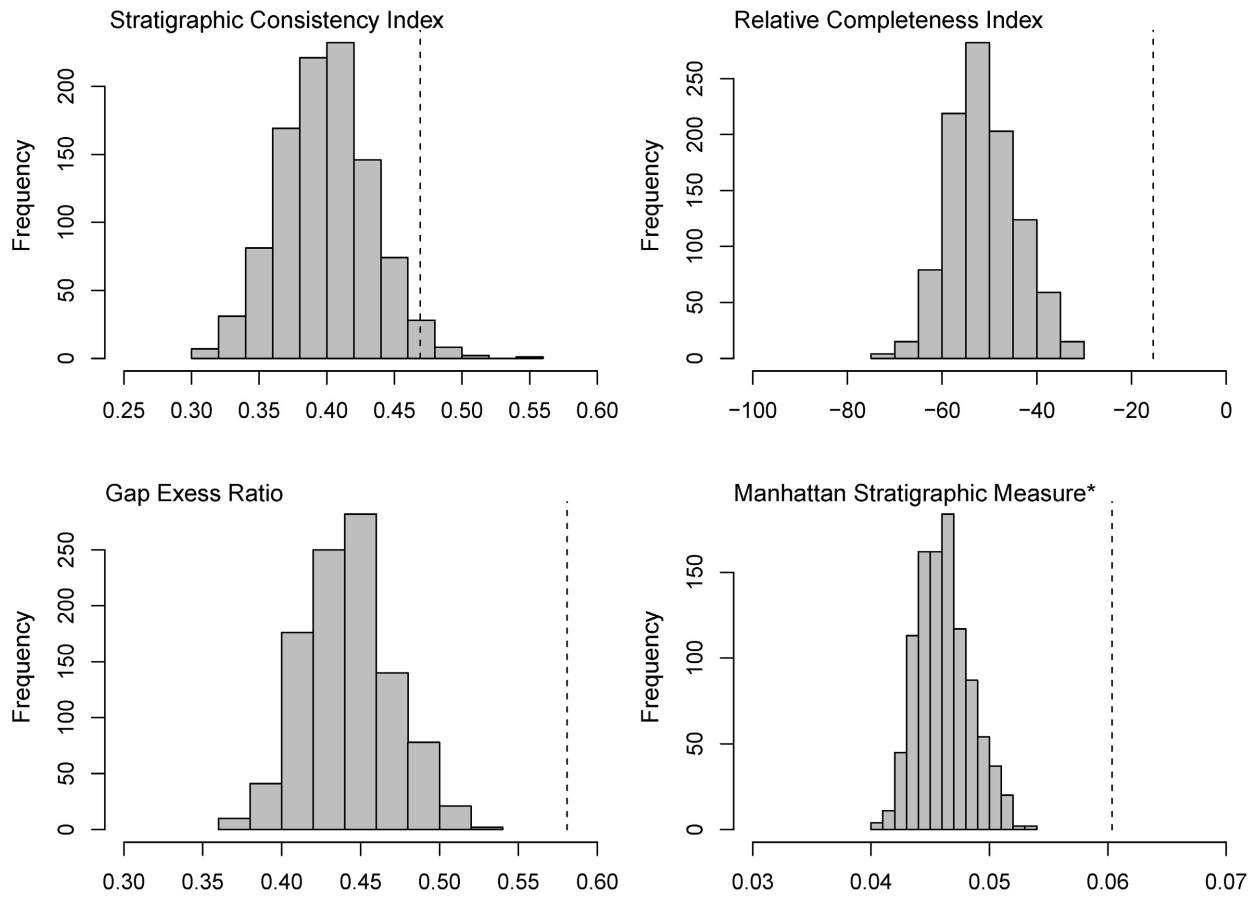


FIG. 4. Stratigraphic congruence metrics for the consensus tree recovered from parsimony analysis. For each measure, the observed value (dashed line) is compared to a null distribution produced using a Monte Carlo simulation of 1000 trees.

a sharp decrease in diversity during the late Silurian followed by a plateau during the Devonian, the phylogenetic diversity estimate infers a steady decline throughout this interval.

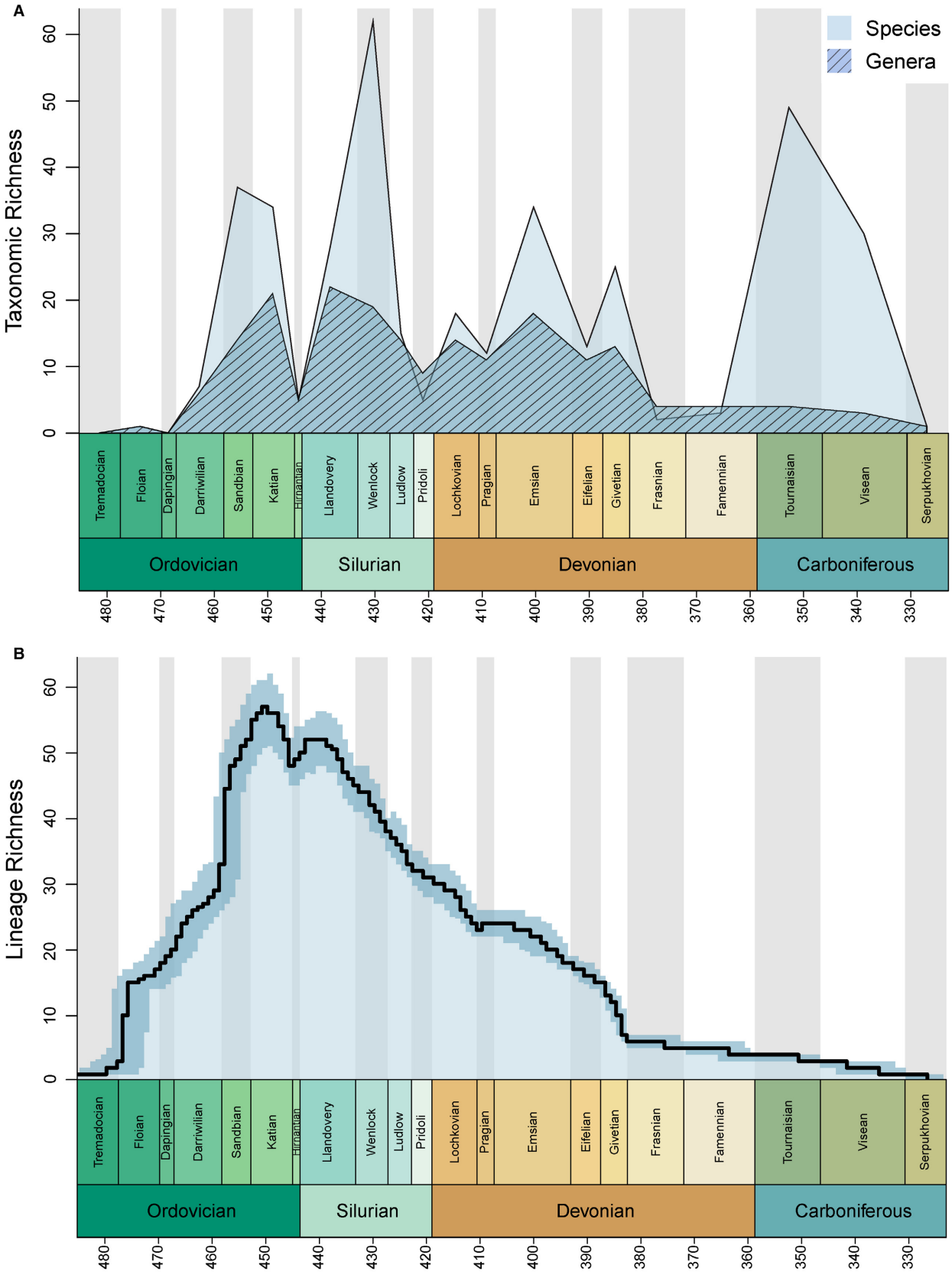
Because only a genus-level tree is currently available for diplobathrids, phylogenetic estimates of diplobathrid diversity at the species level were not possible. Application of the range-through approach to raw counts of sampled species produces diversity patterns that broadly covary with those at the genus level. For both genera and species, all major decreases in diversity appear synchronous, and most diversity peaks correspond closely. Some of the most prominent diversity peaks, however, are offset by one stage in places, and the direction of offset is not consistent. For example, a peak in diversity occurs at both the species and genus level during the Ordovician, but this peak is reached during the Sandbian at the species level and during the Katian at the genus level. By contrast, peak Silurian diversity is reached during the Llandovery at the genus level, but during the Wenlock at the species level. Finally, a notable decoupling between species and genus-level diversification patterns occurs during the Carboniferous when genus diversity gradually decreases, yet species diversity dramatically increases.

DISCUSSION

Implications for systematic revision

The trees presented here provide a more comprehensive view of order Diplobathrida through its geological history

FIG. 5. Diversification history of diplobathrid crinoids. A, genus and species diversity calculated using the range-through method; intervals where species diversity appears lower than genus diversity are an artifact of the range-through method (e.g. instances where no species were sampled for a genus within a time bin, but other species of the genus were found in both preceding and following time bins). B, phylogeny-based estimates of genus-level lineage diversity calculated from a sample of 150 time-scaled trees; solid line represents median diversity, shaded area represents 95% quantiles. Colour online.


A
Taxonomic Richness
Species
Genera
0
10
20
30
40
50
60
Tremadocian
Floian
Dapingian
Darriwilian
Sandbian
Katian
Hirnantian
Llandovery
Wenlock
Ludlow
Pridoli
Lochkovian
Pragian
Emsian
Eifelian
Givetian
Frasnian
Famennian
Tournaisian
Visean
Serpukhovian
Ordovician
Silurian
Devonian
Carboniferous
480
470
460
450
440
430
420
410
400
390
380
370
360
350
340
330
B
Lineage Richness
0
10
20
30
40
50
60
Tremadocian
Floian
Dapingian
Darriwilian
Sandbian
Katian
Hirnantian
Llandovery
Wenlock
Ludlow
Pridoli
Lochkovian
Pragian
Emsian
Eifelian
Givetian
Frasnian
Famennian
Tournaisian
Visean
Serpukhovian
Ordovician
Silurian
Devonian
Carboniferous
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than was previously available, and thus serve as a suitable framework for addressing implications for systematic revision of the clade, particularly at the levels of suborder, superfamily and family. Here, I highlight some of the most significant revisions that should be made based on the results of the present analysis. Formal systematic revisions, however, are left to a later paper that will be dedicated solely to that extensive undertaking. A thorough systematic revision will require reassignment of genera, splitting and combining higher taxa, emending diagnoses, and erecting new families and other higher taxa.

The traditional classification of diplobathrids is largely incompatible with the recovered trees at and above the family level. Although broad separation of diplobathrids into two clades (Clade A and Clade B; Fig. 2) suggests there may be grounds for a two-part division at the level of suborder, this division does not correspond to the suborders Eudiplobathrina and Zygodiplobathrina (Ubaghs 1953), which were previously proposed for Diplobathrida but have long been questioned (Brower 1975; Kelly *et al.* 1978; Haugh 1979; Kolata 1982; Ausich 1986). Defining suborders that correspond to these recovered clades should be undertaken with caution, however, as little to no support was recovered for these groups, and corresponding character suites that could be used for diagnosis are lacking.

Similarly, the traditional superfamily divisions are not recovered as monophyletic groups. Nyctocrinitoidea is not supported as a separate superfamily, as its sole genus is deeply nested within a group of rhodocrinitoids rather than as an isolated, phylogenetically unique lineage. Rhodocrinitoidea is recovered as a paraphyletic group, and the dimerocrinitoids are nested within the rhodocrinitoids. The overwhelming majority of dimerocrinitoids are clustered to form an evolutionary grade, but because most fall along a polytomy, there is some ambiguity as to whether a majority subset might represent a monophyletic group. With subsequent analyses to more thoroughly assess the monophyly of Dimerocrinitoidea (e.g. through sampling of additional species), it may be possible that this superfamily can be retained with some revision. The rhodocrinitoids, however, are split into three groups: Clade A, Clade B1 and Clade B2 (Fig. 2). A number of rhodocrinitoids are also positioned more stemward in the tree below the division between Clade A and Clade B. Revisions to the classification of diplobathrids at the superfamily level should at minimum reflect this three-part division of rhodocrinitoids, separation of the rhodocrinitoid stem lineages and dimerocrinitoid-grade diplobathrids, and demote Nyctocrinitoidea to a lower taxonomic rank. Although support metrics are not recovered for these groupings, many are united by a number of characters that could be used for diagnosis. Typically, characters that are useful for diagnosing clades are conserved across the tree, whereas characters that are more plastic and/or evolve convergently in different lineages are less useful. Some of these characters that may be useful for diagnosis are given in Cole (2018, suppl. table 3). Not all are strict synapomorphies, but nonetheless may be useful for diagnosing the common morphological features shared by members of these groups. Interestingly, the characters listed are overwhelming related to aspects of calyx construction, with only one character reflecting stem morphology and two reflecting arm morphology, which may be indicative of a broader pattern of increased conservativism among calyx characters compared to arm and stem characters.

At the family level, traditional groupings do not correspond directly to recovered clades with the exception of Elpidocrinidae. As a result, the most extensive revisions will be required at the family level via reassignment of many genera. In many families, however, either the majority of their constituent genera form clades (e.g. Emperocrinidae, Opsiocrinidae) or they are closely related (e.g. Pterinocrinidae, Anthemocrinidae, Anthracocrinidae), which will allow many of the pre-existing family names to be preserved during revision. Potential revisions at the family level include: dissolution of Aishacrinidae and reassignment of *Aishacrinus* to the Anthemocrinidae; reevaluation of the genera belonging to Anthracocrinidae; possible elevation of subfamily Gilbertsocrininae to a family; and revision of Opsiocrinidae to remove *Hollowaycrinus* and potentially include other genera in the family. In contrast to some of the higher taxonomic ranks, many of these potential families have high support metrics and numerous characters that can be used for diagnosis. Thus, it is likely that families identified from the tree presented here will be more straightforward to delineate than some of the higher taxonomic ranks.

Many of the characters used to diagnose higher taxa in the traditional classification scheme are not useful for diagnosing the clades recovered here. For example, the arrangement of radials and basals in a single circlet of 10 plates, which was used to separate Zygodiplobathrina from Eudiplobathrina, does not correspond to any recovered clades. Likewise, the distinction between rhodocrinitoid, dimerocrinitoid and nyctocrinitoid conditions, which are related to the separation of the radial plates, are not strictly diagnostic for any monophyletic groups (Fig. 1). Rhodocrinitoid and dimerocrinitoid arrangements of the radial plates have both evolved multiple times, and the nyctocrinitoid condition appears to have been derived from the rhodocrinitoid configuration. However, the rhodocrinitoid and dimerocrinitoid configurations of the radial circlet are generally conserved within clades and may be useful for diagnosis if used in conjunction with other character data.

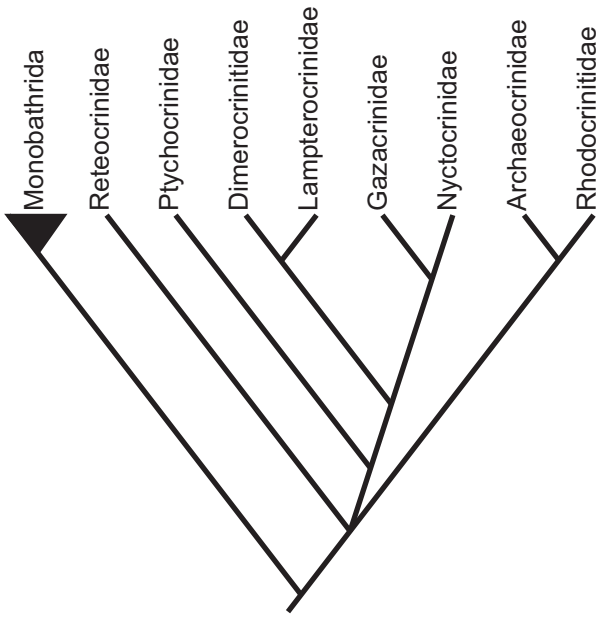


FIG. 6. Summary of relationships among diplobathrid families as proposed by Moore & Laudon (1943).

Interestingly, the topology recovered here is largely consistent with the hypothesis of family-level relationships suggested by Moore & Laudon (1943). Their cladogram depicts three primary diplobathrid groups: (1) reteocrinids; (2) rhodocrinitids and archaeocrinids; and (3) ptychocrinids, lampterocrinids, dimerocrinitids, gazacrinids and nyctocrinids (Fig. 6). Although the reteocrinids have now been reclassified as stem eucamerates and this study has shown that the families themselves may not be cohesive as currently described, the relationships among type genera of most families correspond to those postulated by Moore & Laudon (1943). Similarities include the close relationships between rhodocrinitids and 'archaeocrinids,' between gazacrinids and nyctocrinids, and between dimerocrinitids and lampterocrinids.

Evolutionary history of Diplobathrida

Morphological trends. The analyses presented here do not support the traditional suborder divisions of rhodocrinitoids, dimerocrinitoids and nyctocrinoids on the basis of their radial circlet condition (Fig. 1). Instead, results indicate both rhodocrinitoids and dimerocrinitoids are evolutionary grades rather than clades, and that nyctocrinoids are probably derived from a group within the rhodocrinitoid grade (Fig. 2). The recovered trees imply taxa with rhodocrinitoid-grade morphology (i.e. Clade A; Fig. 2) were the ancestral form and the dimerocrinitoid condition was secondarily derived. This is consistent with less-comprehensive phylogenetic analyses that were previously conducted (Cole 2017). The evolution of dimerocrinitoids from rhodocrinitoids is also temporally consistent with the stratigraphic records of both groups, because the first known appearance of rhodocrinitoids is Floian (Lower Ordovician), whereas dimerocrinitoids do not appear until the Katian (Upper Ordovician) (Fig. 3).

Unsurprisingly, the rhodocrinitoid-grade groups (Clades A, B1 and B2; Fig. 2) share the same morphological characters traditionally used to define the group (Cole 2018, suppl. table 3). However, the three groups exhibit differences in some of the other characters common to most of their constituent genera. Notably, Clade B2 seems to represent more derived forms of rhodocrinitoid-grade diplobathrids characterized by a general trend towards reduction in the number of plates (posterior and regular interrays, interambulacral areas, etc.) and the acquisition of perfect pentameral symmetry through loss of a differentiated posterior interray. By contrast, most members of Clades A and B1 retain complex calyces with more numerous plates and a differentiated posterior interray.

Taxonomic diversification. Although sampled genus diversity is higher for the Silurian than for the Katian, phylogenetic estimates suggest that true diversity was higher during the Katian (Fig. 5). Based on the stratigraphic distribution of Clades A and B, the Katian peak in diversity is predominantly driven by the Clade A rhodocrinitoids, whereas the early to middle Silurian peak in diversity is driven by the dimerocrinitoids and other members of the rhodocrinitoid grade belonging to Clade B. A small subset of taxa from Clade A, however, persisted into the Devonian and Carboniferous and continued to contribute to total diplobathrid diversity through the Mississippian (Fig. 3).

The early radiation of diplobathrids during the Middle to Late Ordovician coincides with the Great Ordovician Biodiversification Event (Webby *et al.* 2004) (Fig. 5B). The GOBE is most notably characterized by rapid increases in taxonomic diversity of many groups of marine invertebrates, particularly at the ordinal and family levels, which resulted in the rise to dominance of the Palaeozoic Evolutionary Fauna (Sepkoski 1981; Webby *et al.* 2004). Recent work on taxonomic diversification of flexible crinoids found diversification patterns similar to those recovered for diplobathrids, with peak taxonomic diversity being reached during the Katian, suggesting this pattern applies to Ordovician crinoids as a whole (Wright & Toom 2017).

Diversity tabulated from range-through data suggests that diplobathrids were severely impacted by the end-Ordovician extinction, but phylogeny-based estimates indicate a large number of lineages survived the event and radiated during the Llandovery (early Silurian) (Fig. 5). Although sampled genus diversity is slightly higher for the Llandovery than the Katian, phylogeny-

based diversity estimates suggest this is a sampling artifact and true genus richness was higher during the Katian. Although phylogenetic estimates of species diversity are not available, range-through species diversity is notably elevated during the early Silurian, which may represent increased species-level origination relative to the origination of new genera.

Following the recovery of diplobathrids during the early Silurian, diplobathrid genus diversity steadily declined through the Silurian and Devonian (Fig. 5B). Although diplobathrids experienced a sharp drop in diversity during the Devonian, it did not correspond to either of the recognized extinction pulses that comprise the late-Devonian mass extinction: the Kelwasser Event at the Frasnian–Famennian boundary and the end-Devonian Hangenberg Event (Raup & Sepkoski 1982; McGhee 1996; Caplan & Bustin 1999; Bambach 2006). Instead, this sharp drop in diversity occurred earlier at the Givetian–Frasnian boundary, which has been recognized as a notable period of extinction among crinoids (Baumiller 1994).

At the genus level, diplobathrids never recovered from these major losses in diversity. Despite near-constant genus diversity through the Late Devonian and Mississippian, however, a dramatic increase in species-level diversity occurred (Fig. 5A). Sampled species-diversity rivals that of the Llandovery and exceeds that of the Katian, yet the origination of new genera lagged far behind that of species, with only a single new genus, *Yunnanocrinus*, appearing during the Mississippian. The remaining genera responsible for this pattern of low genus diversity and high species diversity were *Rhodocrinites*, *Gilbertsocrinus* and *Cribanocrinus*. Although it is possible that undersplitting of these genera may be partly responsible for the unusually high diversity of species relative to that of genera, this may also represent a truly asynchronous pattern of diversification during the Mississippian when diversification at the species level dramatically outpaced that at the genus level. One explanation is that this pattern might be the result of increased niche partitioning during the Mississippian, when crinoids as a whole reached the highest diversity in their evolutionary history during the 'Age of Crinoids' (Kammer & Ausich 2006). Increases in relative abundance and density of organisms within communities can drive ecological speciation through competition and resource partitioning, which may be the case for Mississippian diplobathrids (McKinnon *et al.* 2004; Schluter 2009; Nosil 2012). Given the species-area effect and vast geographical expanses of carbonate ramp habitat that were available during this interval (Kammer & Ausich 2006), Mississippian diplobathrids may exemplify a Type III–style diversification (*sensu* Jablonski 2017), where taxonomic diversification outpaces increases in morphological disparity. Further evaluation of Mississippian diplobathrid genera for potential undersplitting and construction of phylogeny-based diversity curves at the species level may help to assess the validity of this pattern and identify what underlying processes might be responsible for its creation.

CONCLUSION

Here, I have presented the most comprehensive phylogenetic analysis of diplobathrid crinoids to date and investigated their diversification history using both taxonomic and phylogenetic estimates of diversity through time. Recovered trees suggest that significant systematic revisions will be required for the classification of diplobathrids to adequately reflect the evolutionary history of the group. In particular, suborders Eudiplobathrina and Zygodiplobathrina are not supported, which is consistent with the expectations of previous workers; superfamily Nyctocrinoidea is not supported; and superfamilies Rhodocrinitoidea and Dimerocrinitoidea are recovered as paraphyletic grades. Results indicate that Dimerocrinitoidea was derived from rhodocrinitoid stock, and was ancestral to a group that secondarily evolved rhodocrinitoid-grade morphology. Many of the recovered groups that are more nested in the tree, which broadly correspond to families, are well-supported by both support metrics and character data, which can be used for future phylogeny-based revisions to the classification of crinoids. Revisions to the higher classification of diplobathrids may require further phylogenetic investigations of subclades to recover groups that are well-supported by both phylogenetic support metrics and morphological character suites that can be used for diagnosis. However, the results presented here suggest that characters relating to the crinoid calyx are probably more informative for diagnosing clades than are arm and stem characters.

Phylogenetic estimates of diplobathrid diversity provide more realistic measures than range-through counts of sampled taxa by accounting for unsampled lineages. Notable events in the diversification history of diplobathrids include: (1) peak genus diversity was reached during the Katian, which is consistent with results of diversity analyses of other crinoids and is coincident with the GOBE; (2) following the end-Ordovician extinction, diplobathrids radiated during the early Silurian (Llandovery) but their generic diversity did not reach that of the Katian; (3) diplobathrid diversity steadily decreased through the middle–late Silurian and Devonian, with a sharp drop in diversity occurring at the end-Givetian; (4) genus diversity remained very low until the ultimate extinction of diplobathrids at the end of the Mississippian; and (5) species diversity increased dramatically during the early Mississippian, which may represent increased ecological speciation during the 'Age of Crinoids'.

The results of this study provide a necessary framework for many subsequent studies. It is a major step towards both a phylogenetically-informed revision of Diplobathrida and construction of a comprehensive phylogeny for crinoids. In addition, the inferred trees can be further utilized in phylogeny-based investigations of macroevolutionary patterns through the Palaeozoic.

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DATA ARCHIVING STATEMENT

Data for this study (including supplementary tables and figures, matrices and constraint tree used for parsimony and Bayesian phylogenetic analyses, and trees recovered from phylogenetic analyses) are available in the Dryad Digital Repository: https://doi.org/10.5061/dryad.54q3f36

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